STARWEAVER

Non-Graded Project: AI Agent Architecture Design for Enterprise SOC Integration

Objective

Design and document a comprehensive AI agent architecture for a mid-sized enterprise's Security Operations Center (SOC), demonstrating understanding of AI agent fundamentals, security considerations, operational implementation, and ethical governance frameworks learned throughout the course.



Project Description

You are a cybersecurity consultant tasked with proposing an AI agent solution for TechCorp, a mid-sized technology company with 500 employees. The company currently operates a traditional SOC with 3 security analysts working 24/7 shifts. They are experiencing alert fatigue (200+ alerts daily), with 85% being false positives. The company wants to implement AI agents to enhance their cybersecurity operations while maintaining human oversight and addressing ethical concerns.

Step 1: AI Agent Architecture Analysis. Using your knowledge from Module 1, design the core architecture of an AI agent system for TechCorp's SOC. Document:

- The role of LLMs as the "brain" of your AI agent
- Key components needed (perception, reasoning, memory, action and some other modules)

- How the agent will interact with existing SOC tools and workflows
- Data types the agent will process (logs, alerts, threat intelligence or others)

Step 2: Threat Detection and Analysis Capabilities

Based on Module 2 concepts, define specific detection and analysis capabilities your AI agent will provide:

- Real-time threat detection scenarios the agent will handle
- Types of vulnerabilities it will automatically identify

- Network security monitoring capabilities
- Log analysis and anomaly detection processes

Step 3: Incident Response and SOC Integration

Drawing from Module 3, outline how your AI agent will enhance incident response and SOC operations:

- Automated response workflows the agent will execute
- Human-AI collaboration model for the SOC team

- Integration with existing security tools and playbooks
- Proactive defense capabilities and threat hunting support

Step 4: Security and Ethical Framework

Using Module 4 knowledge, address security risks and ethical considerations:

- Identify 3 major security vulnerabilities specific to your AI agent design
- Propose mitigation strategies for prompt injection and data poisoning attacks
- Design a governance framework ensuring responsible AI deployment

Outline human-in-the-loop (HITL) mechanisms for critical decisions

Step 5: Implementation Roadmap and Future Considerations

Create a high-level implementation plan considering:

- Phased deployment approach (pilot, expansion, full integration)
- Success metrics and KPIs for measuring AI agent effectiveness
- Future enhancements and emerging AI technologies to consider
- Training requirements for SOC analysts working with AI agents

Step 6: Compile the Report

Organize information following this structure for peer review:

1. **Title Page:** Project title, your name, date.
2. **AI Agent Architecture Design:** Detailed architecture overview with LLM integration, core components, and data flow description.
3. **Cybersecurity Capabilities Matrix:** Comprehensive list of detection, analysis, and response capabilities with specific use cases.
4. **SOC Integration and Human-AI Collaboration Model:** Workflow integration, collaboration framework, and operational procedures.

5 **Security Risk Assessment and Ethical Governance:** Identified vulnerabilities, mitigation strategies, governance framework, and HITL mechanisms.

6 **Implementation Roadmap:** Phased deployment plan, success metrics, future considerations, and training requirements.

7 **Recommendations:** Two actionable recommendations for TechCorp's AI agent implementation.

By completing this project, you will demonstrate comprehensive understanding of AI agents in cybersecurity, from technical architecture to ethical deployment, preparing you for real-world implementation decisions in cybersecurity environments.

Instructions for Documenting and Sharing The Project for Peer Review

Document the Project

- Use word processing software to create your report.
- Ensure all sections of the project document structure are completed.
- Proofread and edit your report for clarity and accuracy.

Share the Project

Document: Save the report in PDF format.

Upload: Share in the “Peer Review” area of the course shell with a brief description.

Peer Review

Review the projects submitted by your peers.

Provide constructive feedback and comments on their work.